

Advanced Learning for Phishing URLs Detection to Secure Consumer-Centric Applications

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Abstract—This research aims to develop a machine learning-based phishing attack detection framework. Phishing attacks have become one of the most prevalent cybersecurity threats, potentially compromising sensitive information such as login credentials, financial details, and personal data. A machine learning (ML)-based approach for phishing URL detection can help improve the effectiveness and efficiency of phishing detection. The healthcare industry received multiple attacks recently, including phishing attacks. There may be multiple reasons behind the attacks, including outdated resources and weak security mechanisms. The proposed ML-based approach trained on a large dataset to learn the characteristics that distinguish Phishing and Non-phishing URLs to prevent the healthcare industry from phishing attacks. The features were taken from the URLs and provided to the model for training. By detecting phishing URLs in real time, individuals and organizations can take proactive measures to protect themselves from the damaging effects of phishing attacks. The proposed ML-based model detected the phishing URL with 99.00% accuracy, indicating that most attacks were detected.

Index Terms—Machine learning, Phishing, cybersecurity, cyberattack, consumer-centric application.

I. INTRODUCTION

DATA security employs protective digital privacy measures to prevent unauthorized access to computers, databases, and websites.¹ Many organizations have been affected by data breaches and cyberattacks to date [1], [2]. Organized crime groups have actively exploited the following flaws of the healthcare system in recent years: (i) Endpoints for Remote Desktop Protocol (RDP) or Virtual Desktop. Web server configurations that are not safe, system administration, and Electronic Health Record software that is not secure. (ii) Authentication procedures that are either trivial or weak, such as weak passwords or faulty authentication implementations. Vulnerabilities in remote connectivity software. (iii) Platforms

no longer maintained, such as Windows Server 2003 and Windows Server 2008, have weaker security due to poor password selection.

A phishing attack is a type of cyberattack where attackers use social engineering tactics to trick users into disclosing sensitive information, such as login credentials or financial information [3]. Phishing attacks have become increasingly sophisticated in recent years, making them harder to detect and mitigate [4]. Recent research confirm the effectiveness of advanced methods such as Deep Learning (DL), Blockchain, Reinforcement Learning, and Machine Learning (ML) in mitigating security threats like intrusion detection, distributed denial of service (DDoS), and phishing attacks in automated vehicles [5], [6] and IoT networks [7]. To mitigate the DDoS and intrusion attacks on IoT networks, researchers use ML, reinforcement learning, and DL [7], [8], [9] for improving network security by identifying cyberattacks. This research mainly focused on the preventive measures from phishing attacks on the healthcare industry with machine and deep learning techniques. There have been several breakthroughs in the future accessibility of more remote healthcare institutions because of recent technological advancements. Healthcare technologies are moving away from local clinics to monitor patients with smartphones, sensor-based patient-monitoring systems, and other healthcare technologies [10]. Among other fields, advanced AI technologies have benefited patient safety, chronic disease management, and drug supply chain security. By checking compliance with hospital sanitation requirements, automating activities, and analyzing data, cameras and sensors can aid patient safety and monitoring [11].

During the COVID-19 outbreak, private and public organizations received multiple attacks to access confidential information [12]. The attacker mainly aims to steal the consumer's data such as security numbers, credit card details, passports, biometric data, and medical history information [13]. Many unauthorized changes in the healthcare database—such as prescriptions of the wrong medication and medical treatment, were reported. The system was broken down, the scheduled operations were cancelled, and the patient diagnosis and diagnostic procedure were altered. Incorporating cutting-edge technology into the healthcare sector has boosted healthcare precision while raising demand for advanced security [14], [15]. Because of its quantity of valuable data and insufficient security posture, healthcare is one of the most targeted sectors by attackers constantly confronted with emerging cyber threats. According to the SANS cyber threat report, 94% of healthcare organizations reported cyber-attacks [16]. Still,

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¹<https://www.ibm.com/topics/data-security>

only a small percentage of the total number was reported in the public domain, leaving the rest unidentified or unreported [17].

Cyber attacks on the healthcare industry can be global or state-sponsored and are motivated by many influential factors [17], [18]. The perpetrators leaked medical files of prominent athletes in a recent hack against the World Anti-Doping Agency,² but their motive remains unknown. Apart from exploiting technology flaws, these attacks also target human aspects, the weakest link in cybersecurity. As a result, effective cybersecurity must become ingrained in healthcare organisations. The purpose of this research is to find answers to the following questions: (i) Why is the health industry being targeted by attackers? (ii) How did the healthcare sector become vulnerable? (iii) What are the vulnerabilities in the healthcare industry, and how may they be avoided? (iv) What security measures can be implemented to improve the confidentiality of healthcare? And provide the solution to secure the medical record on the cloud efficiently.

The health industry is a prime target for attackers due to the vast and valuable troves of sensitive data it holds, encompassing personal and medical information. This data is highly sought after for identity theft, insurance fraud, and other illicit activities [4], [12]. Failing to detect phishing URLs in healthcare services and adequately protecting sensitive data can have severe consequences. It may lead to a significant data breach, compromising the confidentiality of patient records and personal information. Operational disruptions and compromised patient safety may occur as unauthorized access to healthcare systems disrupt services. Additionally, the compromised security posture could invite further cyber threats, posing risks to the overall integrity of the healthcare organization's network [13]. To prevent these dire outcomes, it is imperative for healthcare services to prioritize the detection of phishing URLs, implement robust cybersecurity measures, and foster a culture of security awareness among staff. These observations indicate the requirements for phishing attack detection in the healthcare industry. The major contributions of this research include the following:

- 1) A new intelligence system for healthcare cybersecurity is proposed in this research. The proposed model helps prevent the healthcare industry from phishing attacks, which utilizes the processing of the incoming URLs and detects whether it is phishing or not.
- 2) The proposed method mainly uses URL-based features for model training, including domain names, length of the URLs, and other relative information, which helps in phishing URL detection.
- 3) The performance of the proposed method is evaluated with a large dataset having more than four lakhs of URLs. The model achieved 99% accuracy for Phishing URL detection with extracted features.

The rest of the paper is organized as follows: Section II discusses the related works. Section IV discusses the working of the proposed low-complexity model for securing files and Convolutional Neural Network for Phishing URL prediction.

Section V discusses the findings, and finally, Section VI concludes this work.

II. LITERATURE REVIEW

Phishing attacks are common nowadays, and worldwide research has focused on solving this issue. Phishing is a common attack where the victims believe in the incoming message and get trapped. This problem belongs to the classification task, and hence, the metrics like precision, recall, F1-score, and accuracy [19], [20] were mostly used by the researchers for their model's performance evaluation. This section highlights some relevant research on phishing attack detection using machine and deep learning techniques [21], [22]. A detailed study reported in recent review research works about different approaches to detect phishing attacks with the help of different AI-enabled techniques like ML, DL, and hybrid learning. They also surveyed used dataset and ML algorithms comparative performance [23], [24]. Catal et al. [25] presented a summary of research on Phishing detection with deep learning techniques. Hybrid algorithms and DNN performed better than other deep learning techniques. Notably, 72% of the research did not include feature selection methods when building their models. The most often used dataset was found to be PhishTank. The two most popular deep learning frameworks are Keras and Tensorflow, although 46% of publications did not name a framework.

A phishing attack is preferable to capture the consumer's (user's) sensitive information; the attacker clones the actual identity, like a website; hence, most of the time, it is unrecognizable [26], [27]. In this process, the attacker uses Email communication where a malicious link is shared with victims aiming to steal confidential information like credit card details, authentication details like username and password, or explicitly install malicious software to their local machine. Aljabri and Mirza [28] utilized different combinations of ML and DL models to develop a phishing detection model. They used a composition of domain-based, URL lexical-based, and content-based features. The results said the ML-based random forest classifier provided the best accuracy. ML and DL algorithms are used by Butt et al. [29] to detect phishing attacks with a cloud computing model. They used SVM, NB, and LSTM models to develop their work.

A link to a fake website that looks authentic may be included in a phishing email, asking the recipient to provide sensitive information. Alternatively, the email may contain an attachment that, upon opening, causes the user's device to run malicious code. This code may install malware, including ransomware, Trojans, and Keyloggers, compromising the affected system's security and integrity [30]. A supervised approach for phishing website detection and achieved 94.8% accuracy [31]. Many frameworks are suggested for detecting phishing attacks. However, the workings of the models and extracting features approaches must be conveyed properly. This research fills the gap by providing a detailed description of the machine, deep learning models, and features used to build the framework.

²<https://www.bbc.com/news/world-37369705>, accessed:02-02-2022

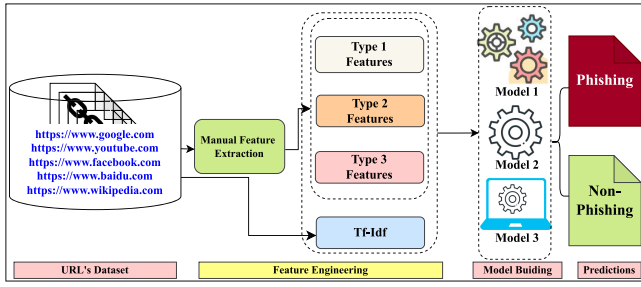


Fig. 1. Framework of detecting Phishing URLs with learning models.

III. PROBLEM STATEMENT

This research aims to detect Phishing URLs to prevent the healthcare industry from cyberattacks. The URLs attached in the email can be manipulated in such a way that a human being can trust them easily and hence click and share their confidential information. To prevent the healthcare industry from such a Phishing attack, a machine learning-based model is suggested in this work. Mathematically, suppose a URL is formed with n number of characters, i.e., $\{c_1, c_2, c_3, \dots, c_n\}$. The number of URL-based features say $\{f_1, f_2, f_3, \dots, f_m\}$, will be extracted from each URL for model development. Here, m represents the number of features. The URL of the database used for the model development will belong to either Phishing (Ph) or Non-Phishing (nPh) category. Mathematically, it can be represented as, $U_i \in C$, where $C = \{Ph, nPh\}$, and $i = 1, 2, 3, \dots, n$. The proposed system aims to detect the URL's (U_i) category, i.e., Ph or nPh .

IV. METHODOLOGY

This research aims to design a Phishing URL detection system. Machine learning and deep learning-based models are trained and tested to identify the best-performing models in several settings. The workflow of the proposed methodology is shown in Fig. 1. The dataset used in this research contains a total of 420,464 pre-processed samples, out of which the Phishing samples are 75,643, and the remaining 344,821 samples are in the Non-Phishing category.

A. Preprocessing and Feature Extractions

The URL dataset contains a lot of noise, which needs proper care during preprocessing. The special characters, like @, _, and others, carry meaningful information; hence, they can not be removed directly. Instead can be used as a feature to represent the URLs. Initially, the parsing steps are performed on URLs to extract different components such as the scheme, domain, path, query string, and fragment identifier, followed by normalization to make them consistent and reduce redundancy. For example, removing unnecessary components such as default ports, fragments, and query parameters that do not affect the page's content. URLs are encoded to handle special characters or non-ASCII characters. Further, it was tokenized to split them into meaningful segments that can be further analyzed or used as features in machine learning models.

Most of the URLs carry only 15-20 characters, so extracting the meaningful features that describe the URLs is challenging.

To build the framework for Phishing detection, three types of features, Type 1, Type 2, and Type 3, are extracted from the URLs in this research, some of the features used in Singh et al. [32] for helpfulness analysis of user's review. The complete list of the extracted features and their explanation are motioned in Table I.

B. Classifiers for Building Predictive Frameworks

1) *Decision Tree*: The Decision Tree (DT) algorithm is a popular machine learning algorithm for classification and regression tasks. It is a tree-like model that breaks down a dataset into smaller subsets while simultaneously, an associated decision tree is incrementally developed. The algorithm involves two key concepts: information gain and entropy.

Entropy measures the impurity of a set of examples. For a given set S , the entropy $E(S)$ is defined as:

$$E(S) = -p_1 \log_2(p_1) - p_2 \log_2(p_2) - \dots - p_c \log_2(p_c) \quad (1)$$

where p_i is the proportion of examples in S that belong to class i , and c is the number of classes. The entropy is maximal when all classes are equally represented in the set and minimal when the set contains only examples of one class.

Information gain is a measure of the usefulness of a feature in splitting the dataset. For a given feature A , the information gain $IG(S, A)$ is defined as the difference between the entropy of the original set S and the weighted average of the entropy of the subsets generated by splitting S on feature A :

$$IG(S, A) = E(S) - \sum_{v \in A} \frac{|S_v|}{|S|} * E(S_v) \quad (2)$$

where S_v is the subset of S for which feature A has the value v , and $|S_v|$ is the number of examples in S .

The algorithm uses these concepts to recursively split the dataset by selecting the feature that maximizes information gain at each node of the tree. The splitting continues until a stopping criterion, such as a maximum depth or a minimum number of examples in a leaf node, is met. Once the tree is trained, it can be used to classify new examples by traversing the tree from the root to a leaf node based on the values of the features of the example. The class label of the leaf node is then assigned to the example.

2) *Logistic Regression*: Logistic Regression is a widely used method for binary classification, which means predicting whether an instance belongs to one of two possible classes. It works by modeling the probability of the binary response variable as a function of one or more predictor variables. It uses a logistic or Sigmoid function, which maps any real-valued input to a value between 0 and 1. The logistic function is defined as:

$$p(y = 1|x) = 1/(1 + \exp(-z)) \quad (3)$$

where $p(y = 1|x)$ is the probability of the positive class given the predictor variables x , z is a linear combination of the predictor variables weighted by their respective coefficients. The linear combination z can be expressed as:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k \quad (4)$$

TABLE I
LIST OF FEATURES AND THEIR EXPLANATIONS

Type	Name	Explanation
Type 1	Length of URL	Length of the URL was calculated using len() function, including special characters
	Length of Hostname	Length of the Hostname was calculated using len() function
	Length of Path	Length of the path was calculated using len() function
	Length of high level domain	Length of the high level domain was calculated using len() function
Type 2	Number of '-'	Total number of underscore character present in the URL is counted
	Number of '@'	Total number of at symbol present in the URL is counted
	Number of '?'	Total number of question mark symbols present in the URL is counted
	Number of '%'	Total number of percentage symbol present in the URL is counted
	Number of '.'	Total number of full stop symbol present in the URL is counted
	Number of '='	Total number of equal to symbol present in the URL is counted
	Number of 'http'	Total number of 'http' string present in the URL is counted
	Number of 'https'	Total number of 'https' string present in the URL is counted
	Number of 'letters'	Total number of 'letters' present in the URL is counted
	Number of 'www'	Total number of 'www' string present in the URL is counted
	Number of Digits	Total number of digits [0-9] present in the URL is counted
Type 3	Use of IP or not	The URL uses IP or not
	Use of Shortening URL or not	The URL is Shorten or not

where β_0 is the intercept or bias term, $\beta_1, \beta_2, \dots, \beta_k$ are the coefficients of the predictor variables $x_1, x_2, \dots, x_k$, respectively.

To train the model, we need to estimate the values of the coefficients that best fit the training data. This is typically done by maximum likelihood estimation, which involves finding the values of the coefficients that maximize the likelihood function of the observed data. The likelihood function measures how well the model fits the training data and is defined as:

$$L(\beta) = \prod [p(y_i = 1|x_i; \beta)]^{y_i} * [1 - p(y_i = 1|x_i; \beta)]^{(1-y_i)} \quad (5)$$

where y_i is the binary response variable for the i th training example, x_i is the vector of predictor variables for the i th example, and β is the vector of coefficients. Once the coefficients of the model are estimated, the model can be used to predict the probability of the positive class for new instances by substituting their predictor variable values into the logistic function.

3) *Random Forest*: It is an ensemble machine learning algorithm that combines multiple decision trees to improve the accuracy and robustness of classification tasks. The algorithm creates many decision trees and combines their predictions through a voting scheme to make the final classification decision. The Random Forest algorithm involves the following steps:

- **Bootstrap Sampling**: Randomly select a subset of the training data with replacement to create multiple datasets of the same size as the original dataset. This process is known as bootstrapping, and the resulting datasets are called bootstrap samples. The bootstrap sampling process involves randomly selecting a subset of the training data with replacement to create multiple datasets of the same size as the original dataset. The probability of selecting any particular example in the bootstrap sample is given by:

$$P(x_i \text{ in sample}) = 1 - (1 - 1/n)^n \quad (6)$$

where n is the size of the original dataset, and x_i is the i th example in the dataset.

- **Feature Selection**: For each bootstrap sample, randomly select a subset of the features for training the decision tree. This process, known as feature selection, helps reduce the correlation between the trees.
- **Decision Tree**: To train a decision tree on each bootstrap sample using the selected features. The decision tree is trained using the recursive partitioning algorithm, which recursively splits the data into subsets based on the value of a single feature in a way that maximizes the information gain or reduces the impurity of the target variable.
- **Ensemble Learning**: Once all decision trees are trained, the predictions are combined through a voting scheme to make the final classification decision. The most popular voting scheme in Random Forest is majority voting, where the class with the most votes across all trees is selected as the final prediction. Let T be the number of decision trees, and $y^{(i)}$ is the predicted class label for sample x_i . Then, the final prediction y of the Random Forest Classifier is given by:

$$y = \operatorname{argmax} \left(\sum_{i=1}^T I(y^{(i)} = j) \right) \quad (7)$$

where j is the class label, and I is the indicator function that returns one if the condition is true and 0 otherwise.

4) *Convolutional Neural Network (CNN)-a Deep Learning Framework*: The CNN framework is used to test the performance of the URL prediction. The working of the CNN model with URLs will be discussed in detail in this section. The CNN network mainly consists of three layers: (a) Embedding, (b) Pooling, and (c) Fully connected dense layer followed by output layer [19]. The embedding layer is used to prepare the sentence matrix (S) by mapping the words of the sentence into vector form. A convolution operation is performed in the next phase of the CNN model. Suppose a sentencing matrix (S) having dimension $n \times d$, and a kernel (k) of bi-gram $2 \times d$, then the number of features extracted by performing convolution operation will be $(n - 2) + 1$. In simple, if the value of n is 50, and a bi-gram kernel $k = 2$ is used for convolution, then $(50 - 2) + 1$, i.e., 49 features

will be produced. The process of convolution operation over the sentence matrix having dimension $d=4$, and the number of words (n) is 4, with the kernel ($k=2$) is discussed below:

$$\mathbf{S} = \begin{bmatrix} wv_{11} & wv_{21} & wv_{31} & wv_{41} \\ wv_{12} & wv_{22} & wv_{32} & wv_{42} \\ wv_{13} & wv_{23} & wv_{33} & wv_{43} \\ wv_{14} & wv_{24} & wv_{34} & wv_{44} \end{bmatrix} \odot \mathbf{k} = \begin{bmatrix} k_{11} & k_{21} \\ k_{12} & k_{22} \\ k_{13} & k_{23} \\ k_{14} & k_{24} \end{bmatrix}$$

where $\odot$ is the convolution operator.

The first feature (say f_1) was calculated by convolving the kernel k over the first two word's vectors of the sentence S , i.e., wv_1 , and wv_2 .

$$f_1 = wv_{11}k_{11} + wv_{12}k_{12} + wv_{13}k_{13} + wv_{14}k_{14} + wv_{21}k_{21} + wv_{22}k_{22} + wv_{23}k_{23} + wv_{24}k_{24}$$

The second feature, i.e., f_2 was evaluated by performing a convolution operation with the next two word's vector (skipping the i^{th} word, and considered $(i+1)^{th}$, $(i+2)^{th}$) of the sentence, i.e., wv_2 , and wv_3 with same kernel k . The outcomes of the f_2 will be:

$$f_2 = wv_{21}k_{11} + wv_{22}k_{12} + wv_{23}k_{13} + wv_{24}k_{14} + wv_{31}k_{21} + wv_{32}k_{22} + wv_{33}k_{23} + wv_{34}k_{24}$$

The third feature f_3 is calculated in a similar fashion, i.e., the kernel k convolves on the next two words (wv_3 , wv_4).

$$f_3 = wv_{31}k_{11} + wv_{32}k_{12} + wv_{33}k_{13} + wv_{34}k_{14} + wv_{41}k_{21} + wv_{42}k_{22} + wv_{43}k_{23} + wv_{44}k_{24}$$

The last word of the sentence is wv_4 , which was used to extract the third feature f_3 . After sliding the window by 1, only wv_4 word vector is present. Hence, further, no convolution operation is possible. This way, convolution operation extracts three features (f_1, f_2 , and f_3) from the input text.

To extract the hidden patterns from the input, the CNN model used several kernels having different sizes. The system automatically assigns the kernels' values, which vary every time. The kernels' different set of values helps extract different patterns from the input. All extracted patterns are not equally important for prediction and need to filter out the important ones—the pooling layer helps to reduce the number of features. Keras supports various pooling techniques, such as average pooling, max pooling, and global average pooling. In average pooling, the average of a window will pool out. For example, if the feature values $f_1=3$, $f_2=4$, $f_3=2$, $f_4=7$, $f_5=4$ then, average pooling will yield the average value of the features, i.e., $(3+4+2+7+4)/5 = 4$.

CNN model uses many such kernels for extracting the features. The model randomly assigns the value of $f_{11}, f_{12}, f_{13}, \dots, f_{1d}, f_{21}, f_{22}, f_{23}, \dots, f_{2d}$. Hence, for different kernels, a different feature value is obtained. Mathematically, the average pooling operation can be explained as:

$$f_i = \text{avg}(f_1, f_2, \dots, f_5) \quad (8)$$

here f_i is the average value pooled from the window of 5 features ($\text{avg}(f_1, f_2, \dots, f_5)$). This way, from a total set of features k having window size w_s , the number of feature pool out will be:

$$N = \left\lfloor \frac{|k'|}{w_s} \right\rfloor \quad (9)$$

where $|k'|$ is the dimension of the feature vector, and w_s is the size of the max-pooling window. This way, pooling operations

from the extracted features yields a total of N features. In other words, if the total number of features in $|k'| = 50$ and the size of the max-pooling window is $w_s = 4$, then total ($N = \lfloor \frac{50}{4} \rfloor = 12$) features will be pooled-out using pooling operation from the feature matrix k' .

The extracted features pass to the fully connected layer at the network's end for further processing. The neurons present at the fully connected layer embedded with an activation function such as ReLU, Leaky ReLU, Sigmoid or similar ones. The ReLU and Leaky ReLU activation function is generally used at the internal level, whereas the output layer, Sigmoid or Softmax activation function, is preferred. To train the model, the Adam optimizer with binary cross-entropy as a loss function is used in the CNN model.

V. RESULTS AND FINDINGS

Extensive experiments were conducted using machine learning and deep learning models to validate and assess their performance on Google Colab, which has 68.40 GB of disc space and 12.69 GB of total RAM available. The models are implemented in the Keras framework and run in Python using TensorFlow as the backend. The dataset is split into a 3:1 ratio for training and testing all machine learning and deep learning models.

Evaluation Metrics: To evaluate the performance of the proposed model, we used precision, recall, F1-score, and an accuracy [33], [34]. For each class (Phishing, Non-Phishing), the evaluation matrix can be defined in Eqs. (10) to (12).

- The precision of Ph/non-Ph URL can be calculated by obtaining the ratio of predicted Ph (say TP)/non-Ph (say TN) URL as Ph/non-Ph URL to total real Ph (TP and FP) /non-Ph URL (TN and FN) in the dataset as given in Eq. (10).
- Recall of Ph /non-Ph URL can be calculated as a ratio of predicted (say TP)/non-Ph (say TN) as Ph/non-Ph URL to total actual Ph (TP and FN)/non-Ph (TN and FP) URL as given in Eq. (11).
- F1-score is the harmonic mean of precision and recall. The mathematical equation for the F1-score can be seen in Eq. (12).

$$\text{Precision} = \frac{TP}{TP + FP} \quad (10)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (11)$$

$$F1 - \text{score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (12)$$

To detect the URL's category, i.e., Phishing (Ph) or Non-Phishing ($non - Ph$), four models experimented in this research, namely, machine learning based Decision Tree (DT), Logistic Regression (LR), Random Forest (RF) and, a deep learning model namely Convolutional Neural Network (CNN). The experimental outcomes of the machine learning models with different settings are presented in Table II. For ML based classifier, three types of manual features, Type 1, Type 2 and Type 3 were extracted from the URLs as listed in Table I. Type 1 feature is related to the length of a domain name, hostname, path, and others. At the same time, Type 2 features

TABLE II
RESULTS OF ML CLASSIFIERS WITH DIFFERENT SETTING

Features Categories	URL's Categories	DT			LR			RF		
		Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
Type 1	<i>Ph</i>	0.89	0.89	0.89	0.83	0.84	0.83	0.91	0.92	0.91
	<i>nPh</i>	0.75	0.78	0.77	0.82	0.80	0.81	0.88	0.89	0.89
Type 2	<i>Ph</i>	0.90	0.92	0.91	0.94	0.93	0.94	0.94	0.96	0.95
	<i>nPh</i>	0.88	0.87	0.88	0.92	0.86	0.90	0.96	0.92	0.94
Type 3	<i>Ph</i>	0.78	0.70	0.74	0.54	0.60	0.57	0.78	0.76	0.77
	<i>nPh</i>	0.60	0.64	0.62	0.40	0.45	0.43	0.70	0.65	0.67
All	<i>Ph</i>	0.96	0.94	0.95	0.96	0.92	0.94	0.96	0.96	0.96
	<i>nPh</i>	0.93	0.92	0.93	0.89	0.94	0.91	0.94	0.96	0.95
Tf-Idf	<i>Ph</i>	0.95	0.88	0.91	0.94	0.96	0.95	0.95	0.97	0.96
	<i>nPh</i>	0.90	0.88	0.89	0.92	0.91	0.91	0.95	0.96	0.96
Manual and Tf-idf	<i>Ph</i>	0.98	0.97	0.98	0.97	0.95	0.96	0.99	1.00	1.00
	<i>nPh</i>	0.96	0.95	0.96	0.96	0.95	0.95	1.00	0.99	0.99

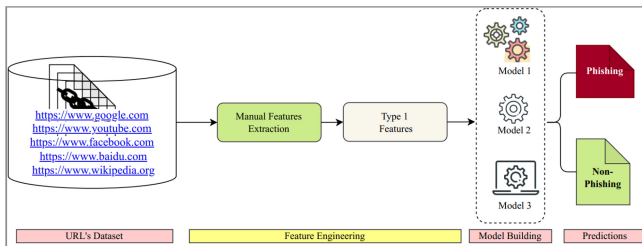


Fig. 2. Framework for model development with Type 1, 2, and Type 3 features.

are the counts of the different characters in the URLs. The third category of the feature (Type 3) is binary, where it checks the presence of the IP addresses in the URLs. The model's performance with each category of the features was shown in Table II.

The ML classifiers DT, LR, and RF were tested with manually extracted feature sets. Firstly, Type 1 features were passed to the models for training purposes and then tested the performance as shown in Fig. 2. The same model (Fig. 2) used for Type 2 and Type 3 features also. For the Type 1 feature, the best-performing model among DT, LR, and RF was the RF classifier where the precision, recall, and F1-score are 0.91, 0.92, 0.91; for Phishing and 0.88, 0.89, and 0.89, respectively, for the Non-Phishing categories.

Now, for Type 2 features, the experiments were repeated with the same classifiers. The performance of the DT classifier is low compared to the LR and RF classifiers. The precision, recall, and F1-score for Phishing categories with LR and RF classifiers are comparable. However, for the Non-Phishing categories, the RF classifier obtained an F1-score of 0.94, whereas the LR classifier obtained 0.90, indicating that the RF classifier is a better choice for the given task. For Type 3 features, the RF classifier yielded a better F1-score than the DT and LR classifiers.

Next, all manual features are used to train the classifiers. With all features, the F1-score of the DT, LR, and RF classifier was 0.95, 0.94, and 0.96 for the Phishing category, indicating improvement of the 1% F1-score value compared to the previously experimented settings. However, the classifier performance with the manually extracted features is limited

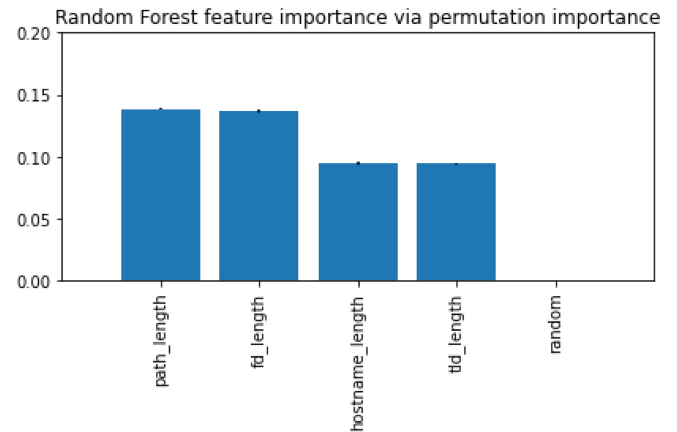


Fig. 3. Plot of the feature (Type 1) importance for random forest classifier.

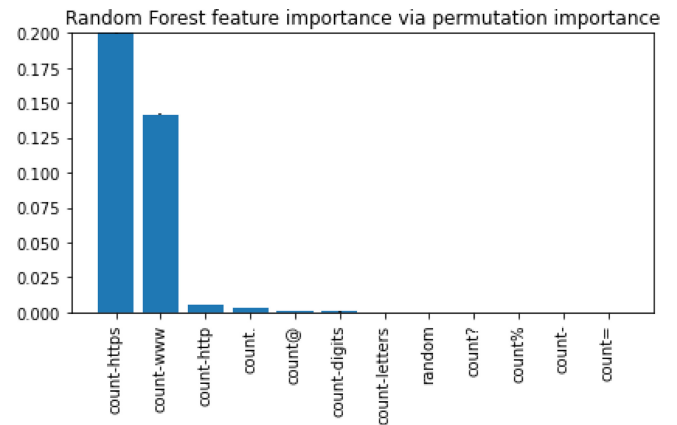


Fig. 4. Plot of the feature (Type 2) importance for random forest classifier.

to a 0.96 F1-score value, indicating a chance of improvement. The manually extracted features performed well in some cases; however, the contributions of the features may vary. To know the contributions of the features, the feature importance graphs plotted as can be seen in Figs. 3, 4 and 5. The contributions of some features are negligible; hence, a new set of features may need to be extracted to get a better prediction performance.

Extracting a new set of manual features may also not guarantee better performance. Therefore, we have used an

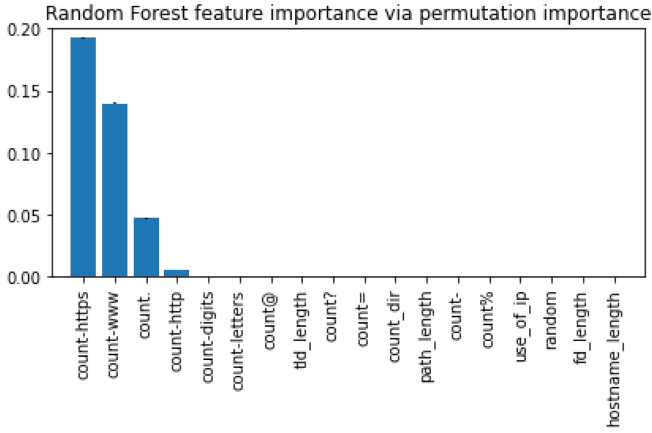


Fig. 5. Plot of the feature (all) importance for Random Forest classifier.

TABLE III
PERFORMANCE OBTAINED WITH CONVOLUTIONAL NEURAL NETWORKS

URL's Category	Precision	Recall	F1-score
One Layer CNN			
<i>Ph</i>	0.93	0.76	0.84
<i>nPh</i>	0.95	0.99	0.97
Macro avg	0.94	0.87	0.90
Weighted avg	0.95	0.95	0.95
Two Layered CNN			
<i>Ph</i>	0.88	0.84	0.86
<i>nPh</i>	0.97	0.97	0.97
Macro avg	0.92	0.91	0.91
Weighted avg	0.95	0.95	0.95
Three Layered CNN			
<i>Ph</i>	0.94	0.89	0.81
<i>nPh</i>	0.98	0.99	0.98
Macro avg	0.96	0.94	0.95
Weighted avg	0.97	0.97	0.97

automated feature extraction technique termed term-frequency and inverse document frequency (tf-idf) to extract the features. The tf-idf is a popular feature extraction technique widely used in natural language processing research. The outcomes obtained with different classifiers when they trained on tf-idf yielded feature vectors are shown in Table II. The RF classifier obtained a 0.96 F1-score for both categories, similar to the Type 2 feature-based outcomes. Hence, it can be said that the tf-idf-based feature reduces the overhead of the manual process of extracting features but is not helpful in improving prediction accuracy for the given problem.

The CNN model also experimented with different settings to improve the phishing URL detection accuracy. In detail, three different settings of the CNN models were experimented with, including one-layered CNN, two-layered CNN, and three-layered CNN. As shown in Table III, the Phishing (*Ph*) category has a Recall value 0.76 with a one-layer CNN model. Next, we increased the convolutional layer and implemented a two-layered CNN. The two-layered CNN model achieved the Recall value for the *Ph* category as 0.84, which is 8% higher than the one-layer CNN model. Further, we have extended an additional layer of CNN (three-layered or 3CNN) and achieved a Recall value of 0.89, which is the best among all previous settings. Based on the experimental outcomes with different settings and models as shown in Tables II and III, it can be

said that the manual features and machine learning-based RF classifier is a better choice for the detecting phishing URLs. The F1-score of both categories, i.e., *Ph* and *nPh*, is 0.96, indicating that the RF model can detect most Phishing URLs.

VI. CONCLUSION, LIMITATIONS AND FUTURE SCOPE

Phishing attacks remain a significant threat to consumers and industries alike, and using machine learning-based approaches can help combat this problem. This paper presents a detailed analysis of machine-learning algorithms for detecting phishing URLs, including decision trees, logistic regression, random forest, and convolutional neural networks. To the best of our knowledge, the URL based features like, length, count of characters like @, ?,., https, and others being used first time to detect the phishing URLs in this research. The identified features with ML algorithms helps to improve the model performance. The experimental results showed that the machine learning-based approach outperformed other models with a recall value of 99%. The proposed model have some limitations, including the possibility of false positives and the need for frequent updates to keep up with new phishing tactics. Despite these limitations, the proposed approach provides a promising solution for detecting phishing URLs and protecting consumers and industries from the financial and reputation damage caused by phishing attacks. As phishing attacks continue to evolve, it is essential to continue developing and refining machine learning-based approaches to stay ahead of the attackers.

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